

ADS1002 Final Report

Assignment: Rossman Drugstore Sales

Project Purpose: Forecasting Sales

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1. Introduction and Background Information

1.1 An Executive Summary

Precise sales forecasting is extremely important for the success of retail stores. Accurate sales predictions provide key information in regards to budgeting, inventory management, staffing and promotional planning. In order to respond to customer demand efficiently, sales numbers need to be anticipated in advance so that resources can be optimised effectively and the stores are able to maintain a competitive advantage. Nonetheless, sales forecasting is a difficult task because multiple influencing factors are involved including but not limited to promotions, store assortment, competitor activity, holidays, seasonal patterns and overall economic conditions.

The project responds to these challenges by investigating sales patterns and developing predictive models for Rossmann, a European drugstore chain with over 3,000 stores across seven countries. The main goal is to identify the primary factors which drive up sales, analyse trends and create forecasting models to aid managerial decision making.

The analysis performed revealed that sales were mainly affected by promotions, store type and competitor proximity. Stores running near new competitors faced temporary reductions in sales whereas larger stores or those running promotions consistently performed well. Seasonal and temporal trends were also observed. Sales peaked during high demand periods such as Christmas, Easter and weekends coinciding with holidays. Promotions consistently raised sales with timing and seasonality influencing their effect. Overall, stores with larger assortment types and those with active promotions were more likely to perform better under competitive conditions.

Predictive modelling employed multiple approaches, including Random Forest, Lasso Regression, XG Boost, Linear Regression, Polynomial Regression, Decision Trees, Logistic Regression and ensemble methods. To improve forecast accuracy by utilising the strengths of each method and reduce overfitting, a stacked ensemble model combining several algorithms was used. Additionally, the feature importance analysis in each of the models confirmed that customer volume, store type, promotional activity, timing and competitor distance were the most important factors in predicting sales.

The goal of this project was to show that when data driven analysis is combined with predictive modelling, it results in actionable insights for retail management. By understanding what drives sales and introducing promotions at strategically planned times, Rossmann can optimise performance, plan and allocate resources effectively and be prepared to respond to seasonal and competitive trends.

Therefore, the integration of business expertise with analytical techniques highlights the ability of predictive analytics to support sustainable business decisions and long term growth. Thus, in order to gain a clear competitive advantage and make strategic decisions, retail stores must make good use of their sales data.

1.2 Project Description and Details of the Supplied Data

Project Description

The sales forecasting project is focused on analysing and predicting daily store level sales of Rossmann across the company's European branches. Rossmann has over 3,000 drugstores in seven countries but each is subject to different market conditions, customer demographic and local competition. Because of this operational variety, the project aims to discern how internal and external factors work together to influence store performance.

Realistic data science workflows for retail forecasting were designed for this project. The group began with data extraction and integrations, where datasets containing information about sales, store characteristics, promotions and contextual variables were obtained from websites such as Kaggle and combined into a unified structure.

Afterwards, the dataset was prepared for analysis through data cleaning and wrangling processes. This included handling missing values, adjusting data types, encoding categorical fields and filtering for relevance. More precisely, the project was centred around 20% of stores to ensure analysis focused on locations with sufficient numbers of sales and consistent data coverage.

Following this, basic summary statistics such as mean, standard deviation, minimum and maximum values were computed for each variable and the overall dataset to get an overview of sales variability and store performance distribution. Then, exploratory data analysis was performed to observe structural and temporal patterns. Graphs such as line plots, boxplots and bar charts were utilised to identify daily and seasonal variation, promotional effectiveness and holiday impacts. By visualising the data, trends like sales peaks, declines or recurring cycles were easily identified and allowed for an intuitive grasp on the factors affecting sales.

Furthermore, relationships between variables were examined through correlation analysis. This analysis was utilised to measure the strength and direction of associations between sales and driving variables such as promotional activity, holiday periods and competitor distance. Scatter plots and correlation coefficients were used to provide a mathematical and visual representation of associations, allowing the easy identification of variables which had the strongest influence on sales outcomes. Reliable forecasting models were created because relationships were initially recognised and used to support management decisions.

Before making any complex models, a basic forecasting model was created to forecast sales over a four week period. The baseline model was intentionally made simple to ensure easy interpretation and display how input variables can be used to estimate sales performance.

Each step helped achieve the overall goal of producing meaningful information for Rossmann to improve store operations. The project took raw sales data and converted it into a structured and actionable knowledge base that can inform operational planning, marketing strategies and long term business forecasting for Rossmann.

Details of Supplied Data

The dataset utilised for the Rossmann project combined several sources of information that together described the daily operations and sales performance of the company's stores across Europe. Rather than coming from a single file, the data was provided in the form of multiple related datasets which contained complementary information. They were eventually merged together during the data preparation process to form a single comprehensive analytical dataset that would allow for exploration, visualisation and modelling.

The data was given in standard spreadsheet format (CSV) which made it compatible with Python. By the end of preprocessing, each row represented an individual observation about the store and date whereas each column corresponded to a measurable attribute. The format allowed store level analysis across time and enabled aggregation for broader comparisons.

The core train dataset contained the operation data recorded on a daily basis for each Rossmann store. This included variables like store identification number, date, daily sales figures, number of customers and binary flags indicating whether a store was open or running a promotion on that particular day. This dataset was central to the time series analysis because it encapsulated the behaviour of sales and customer activity over time. The presence of variables like "Open", "Promo", "StateHoliday" and "SchoolHoliday" provided context that was needed to understand the short term fluctuations linked to store closures, promotional periods and holiday effects.

On the other hand, the store dataset had more static, store level information that stayed consistent on a daily basis. It included qualities describing each store's attributes and competitive environment such as store type,

assortment type and the distance from the nearest competitor. Other variables such as “CompetitionOpenSinceMonth”, “CompetitionOpenSinceYear”, “Promo2”, “Promo2SinceWeek”, “Promo2SinceYear”, and “PromoInterval” describe when competitors had entered the market or when continuous promotions starts. These variables also provide contextual information that helps in explaining longer term differences in sales behaviour across locations.

The structure of the two datasets supported both exploratory and predictive goals. The combination of operational, promotional and competitive attributes enabled correlation analysis and regression based forecasting. Because the data was available at daily level, short term impacts caused by promotions or holidays could be isolated while long term seasonality was also examined.

The supplied Rossmann dataset gave a detailed and well rounded overview of store level performance. Both internal business operations and external contextual factors were captured which made the dataset ideal for investigating factors that drove up sales and testing forecasting models. Therefore, the structure, variables and relationships within the dataset enabled the extraction of meaningful insights to guide strategic and operational decision management across Rossmann’s store network.

2. Data Preprocessing

2.1 Pre-processing Before Loading Data into Jupyter

No manual or out-of-Python pre-processing was performed for the datasets used in this project. The raw CSV files for the Rossmann sales and store information were imported directly into Python and processed programmatically. Since the original files were provided in a consistent tabular format with clear headers, no external editing (for example, editing in Excel, column renaming, file splitting...) was necessary prior to Python import.

2.2 Pre-processing and Manipulation of Data in Python

Data ingestion and initial inspection

All project datasets were imported into Pandas Dataframes for analysis. The principal sources were the store-level metadata and the daily sales transaction records. Immediately upon loading, we inspected each DataFrame with `info()` and `describe()` to identify data types. Cardinalities and missing-value patterns. The Date column in the sales table was converted to a proper datetime data type to support time-based feature engineering and merging with other tables.

Cleaning and missing-value treatment

During exploratory checks the store metadata contained missing values in several fields that relate to competition and ongoing promotions. We applied conservative and justifiable imputation strategies consistent with domain meaning:

- **CompetitionDistance**: missing values in the store metadata were imputed with the median competition distance. This choice preserves central tendency and avoids creating extreme artificial distances; it also prevents row removal that could bias the analysis toward stores with complete competitor metadata.
- **CompetitionOpenSinceMonth / CompetitionOpenSinceYear**: missing entries were set to zero to indicate “no known competitor opening data” and were handled downstream via derived flags and age calculations

- **Promo2 fields:** Promo2SinceWeek, Promo2SinceYear, and PromoInterval were filled with neutral values (0 or “None”) where missing so that stores that do not participate in a recurring promo program are represented explicitly rather than as NaNs.

These imputations were done immediately after merging the sales and store tables so that any further joins or feature engineering would operate on a complete table.

Filtering and rationale

Rows where `Open==0` were removed from analysis that model open-day sales. This was a deliberate choice because closed-store rows always have `Sales==0` and therefore do not convey information about patterns and customer demand on open days; retaining the would bias models toward predicting closures instead of open-day sales and inflate the sample size without improving learnable signals.

Feature engineering

We derived several time and competitor features to capture temporal patterns and competitive pressure:

- **Datetime features:** Year, Month, Day, DayOfWeek, and WeekOfYear were created from the Date column. A binary weekend flag and school-holiday indicator were also retained or derived as relevant.
- **Days / Age calculations for competitors:** CompAge (competitor age) was computed as the difference between the observation year and the competitor’s CompetitionOpenSinceYear. Where the competition open year was missing (set to 0), the derived age was filtered/handled so as to avoid spurious negative ages.
- **Competition distance groups:** For EDA and visualization, competition distances were binned into meaningful ranges (e.g., <500m, 500-2000m, 2-5 km, 5-10 km, 10-20 km, >20 km) to reveal non-linear / threshold effects rather than presuming a strictly linear relationship.
- **Average Sales / Customer / Store:** This variable captures how much each customer spends on average in each store, this gives far more meaningful results than average sale / customer across all Rossmann stores and it is the fifth most important feature in the final model (see figure 4.8c).
- **Ratio of Sales with Promo 1 / Store:** This variable measures how much daily sales increase on average with and without promotion 1 occurring, it was calculated by taking the total average daily sales under promotion 1 and dividing it by average daily sales without, this was done for each store and similar to average sales / customer gives far more meaningful results than if this was done for all Rossmann stores in general. This variable was the sixth most impactful in the final model (see figure 4.8c).

Merging and dataset consolidation

The sales table was left-joined with the store metadata on Store to create a master dataset for analysis. The merge order preserved every sales record (left join) while appending store attributes (competition, promo, assortment, store type). After the merge, we again checked for missing values and confirmed the intended imputation had been applied consistently.

Aggregation for targeted EDA and modelling

For the competitor analysis, we focused on a subset of stores to reduce noise and highlight robust patterns. The notebook calculates the mean sales per store and selects the top 20% stores by mean sales to form `comp_df` — the working dataset for competitor analysis. This selection targets stores where sustained sales allow clearer inference about competitor effects and reduces distortion from small-sample low-volume locations.

3. Exploratory Data Analysis (EDA) Summary

3.1 Weekdays and Holidays EDA

Purpose and Scope

This analysis examined how sales varied across weekdays and holiday periods to understand Rossmann's temporal demand patterns. Recognising when demand peaks enables better resource planning, staffing, and promotional scheduling across stores.

Average Sales by Weekday

Figure 3.1a showed the analysis of daily averages revealed a clear cyclical trend. Sales peaked on Mondays and Sundays, suggesting customers tend to restock after weekends or prepare before the start of the week. Saturdays recorded the lowest sales, likely due to shorter trading hours or reduced shopping activity. Tuesday to Friday sales remained relatively stable, reflecting a predictable mid-week cycle. Additionally, Sunday sales showed the widest variation across stores, suggesting that some locations generated exceptionally high sales, others remained closed.

Average Sales by Holiday Type

Figure 3.1b - c showed the comparison of average sales on public holidays, Easter, and Christmas with regular days, and school holidays with non-school holidays. All major state holidays produced noticeable sales increases, while school holidays showed only marginal effects. These results suggest that customers adjust their shopping behaviour mainly around national or religious holidays rather than school breaks.

Weekday x Holiday Interaction

Figure 3.1d showed a heatmap of weekday and holiday combinations highlighting that Friday and Saturday holidays generated the highest average sales, whereas Sunday holidays produced the lowest, likely due to store closures. This shows that the impact of holidays depends not only on type but also on their timing within the week.

Key Insights for Rossmann

1. Peak demand occurs on Mondays, Fridays, Sundays, and public holidays.
2. Saturdays without holidays are the weakest performers.
3. Sunday sales vary widely due to differing store hours and regional opening restrictions.
4. Friday/Saturday holidays amplify demand, while Sunday holidays show minimal effect.
5. Rossmann should increase staffing and stock replenishment on peak days and use midweek and Saturday promotions to stabilise sales.

3.2 Promotions EDA

Purpose and Scope

This analysis investigates the effect of promotional campaigns on Rossmann store sales performance. Using the variables Promo, Promo2, and PromoInterval, the goal is to quantify how much promotions increase daily sales, explore seasonal and monthly variations in promotional impact, and evaluate how promotional effectiveness interacts with store type and time of year. Understanding these dynamics provides actionable insight for optimizing Rossmann's marketing and operational strategies.

Overall Impact of Promotions

Across all stores, promotions have a strong, consistent positive effect on sales. Average daily sales rose from \$5,963 on non-promotional days to \$8,269 during promotions, an uplift of 38.7%. Figure 3.2a shows a clear upward shift in the median and a wider distribution during promotions, indicating that while most promotions raise sales, some drive exceptional spikes. Promotional days also feature more high-value outliers, suggesting that large-scale or seasonal campaigns can generate significant short-term gains.

Monthly and Seasonal Trends

Monthly performance shows clear seasonal variation (Figure 3.2b–d, Table 3.2a). April, June, and July achieved the strongest lifts (above 43%), making them prime months for major campaigns. May and November recorded the weakest response (~30%), possibly due to customer fatigue or competing market activity.

In absolute terms, December showed the largest sales gain (+\$3,172) due to the synergy between holiday demand and promotions. Quarterly trends highlight Q3 (July–September) as the most responsive period (+42.1%), while Q4 (October–December) had the smallest relative lift (+34.9%) as baseline demand was already elevated.

Store Type and Promo Interaction

Promotions increase sales across all store types but with varying intensity. Figure 3.2e

Type B stores have the highest overall sales, while Types A and C show the largest relative lifts, highlighting stronger responses to discounts. Type D stores show limited improvement, suggesting other strategies such as loyalty rewards may be more suitable.

Key Insights & Recommendations

Promotions boost sales by 30–45%, proving their overall effectiveness. Their impact peaks in Q2–Q3, especially from April to July, and weakens slightly in Q4 when baseline demand is already high.

Rossmann should:

1. Concentrate promotions in Q2–Q3 to maximise responsiveness.
2. Diversify Q4 campaigns with loyalty or bundle offers to prevent saturation.
3. Tailor promotional intensity by store type — prioritising discounts for A and C, and alternative incentives for D.
4. Refine campaign frequency to avoid diminishing returns in strong months.
5. Track ROI by month and store type to continuously optimise the promotional calendar.

3.3 Store Type EDA

Purpose and Scope

The analysis investigates whether store type significantly affects sales performance among Rossmann stores. The dataset includes daily sales and customer numbers for all stores, along with their assortment type (A,B,C or D). Most Rossmann stores are type A, followed by type D, and type C, while type B stores are quite rare. Due to this imbalance total sales alone may not capture accurate store performance by store type, so both total and per store sales are taken into consideration.

Sales by Store Type

When examining total sales, it was found that Type A stores generated the highest overall revenue which was primarily because most Rossmann stores were Type A. Type D also contributed significantly to the overall revenue whereas Type C contributed the least and Type B stores, despite having the smallest number of stores, made a noticeable impact.

When average sales per store were analysed, Type B stood out. Despite the presence of very few Type B stores, they had the highest average daily sales and constantly outperformed Types A, C and D which had similar, lower levels. A parallel pattern was also observed in customer numbers, Type B attracted the most customers out of all stores and the other types lagged behind. This trend showed that store type is an essential factor in explaining differences in store performance.

Store Type Performance with Promotions

The best performing store overall is Type B as it has the highest average sales during both promotional periods and non promotional periods. In contrast, store types C and D have consistent low average sales, suggesting that targeted promotional strategies could benefit such underperforming stores.

Key Insights for Rossmann

1. Type B stores outperform others on a per-store basis, despite being limited in number.
2. Total sales figures are greatly affected by the number of stores in each type, per store statistics are essential for fair comparison.
3. Targeted promotions and customer loyalty initiatives could be effective for lower performing stores such as C and D
4. Store type is very important explanatory variable for both sales and customer traffic

Recommendations

To optimise operations and revenue, Rossmann could:

1. Study what makes Type B so successful and replicate those strategies to identify operational, layout or product strategies driving higher sales
2. Boost traffic in Types A, C, and D through promotions or loyalty programs.
3. Allocate resources based on store type ROI, adjusting staffing and inventory budgets according to potential returns.
4. Experiment with converting lower-performing stores toward the Type B model to test whether adopting similar strategies improves sales performance.

3.4 Competitors EDA

Purpose and scope

This analysis investigates the relationship between nearby competitors and store performance. We examined whether competitor proximity and competitor age (i.e., how long a competitor has been open) are associated with changes in average daily sales for Rossmann stores. The analysis used a carefully merged dataset and focused on the top 20% of stores by mean sales to ensure robust comparisons.

Competitor distance distribution

The CompetitionDistance distribution (Figure 3.4a) reveals that most stores have competitors at varying distances: a subset face very close rivals (<500 m), a larger fraction have rivals within a few kilometers, and some stores have the nearest competitor tens of kilometres away. To expose non-linear effects, distances were grouped into bins (<500m, 500-2000m, 2-5 km, 5-10 km, 10-20 km, >20 km) and mean sales were computed per bin. The resulting bar plot shows a clear pattern: stores with very nearby competitors (the smallest distance bin) generally reported lower mean sales, while sales tend to increase as competitor distance grows up to a point. Notably, there appears to be a threshold effect (Figure 3.4b) in the mid to high kilometre ranges (the notebook identifies around 10 km) where the marginal impact of additional distance becomes small.

Competitor age (time since competitor opened)

We computed CompAge as the difference between the observation year and the competitor's reported open year, and grouped stores by competitor age intervals (0-2, 2-5, 5-10, 10+ years). Boxplots (Figure 3.4c) of sales by competitor age show that new competitors (0-2 years) are associated with lower sales distributions for nearby Rossmann stores — consistent with an initial redistribution of customers when a new rival enters the market. Over the longer horizons, the negative effect of competitor presence appears to attenuate as markets stabilize and customers re-establish routines; older competitors (10+ years) correspond to a relatively smaller sales diminution compared with the immediate aftermath of a new entrant.

Interactions with promotions and store type

Competitor effects are not uniform across all stores. The notebook's EDA compares competitors' influence across store types and promotional status. Results indicate that larger-format stores (Store Type A in the dataset) are less affected by close competitors relative to smaller -format stores. Promotions (variable Promo) interact with competitor effects: stores that run promotions tend to sustain higher sales even in the presence of nearby rivals, suggesting that targeted promotions mitigate some competitive pressure. This highlights the importance of modelling interaction terms (competition x store type, competition x promo activity) in any predictive model.

Simple regression to quantify effects

To quantify the relative importance of competitor features, a simple linear regression was fitted on the `com_df` subset using the exploratory variables CompetitionDistance, CompAge, Promo, SchoolHoliday, and DayOfWeek. The regression coefficients (reported in the notebook) indicate a negative coefficient on CompetitionDistance (i.e., closer competitions corresponds to lower sales and a negative effect for very recent competitor openings (low CompAge), while promotion indicators and day-of-week effects are comparatively stronger predictors. Importantly, the magnitude of the competition coefficients is modest relative to promotion and weekday effects, reinforcing the conclusion that competition matters but is one of several drivers.

Key takeaways

- **Competition proximity reduces sales on average**, particularly when a competitor is within a short-to-medium range (the analysis highlights a threshold near ~10 km where marginal effects decline).
- **New competitors have the largest short-term negative impact**; the effect tends to moderate as the competitive environment matures.
- **The competitive effect varies by store type and promotional activity**: large stores and those running promotions are more resilient to nearby competition.
- **Implication for modelling and business**: include competition distance and age as features, and consider interaction terms with promotional activity and store type. Relying on competition features alone would be insufficient; promotions and day-of-week remain strong, predictable drivers of sale

3.5 Weather EDA

Purpose and scope

This analysis explores how weather was used in our final prediction and its impact on the final model. Weather data was collected from the Kaggle competition discussion forums, it consisted of sixteen csv files. A csv file showing each store ID and the German state in which the store is located and another sixteen csv files listing the weather in each state at each date in the dataset. The weather data was then merged with the train store data such that each column (each day for each store) now had its location (state) and all weather variables.

Weather Variables

The complete set of weather variables can be found in the appendix in Figure 3.5a. In short however the variables covered temperature, humidity, wind speed, cloud cover and visibility on each day.

Weather and Sales trend

Graphing monthly temperature and sales over the two year period of the data (Figure 3.5b) demonstrates a clear cyclical relationship between them thus justifying sales' inclusion in the analysis. Evidently during warm weather June - October in Germany, Rossmann sales are low, however as the weather cools and Winter begins in Europe, Rossmann sales spikes. This trend is mirrored in both years of the data, suggesting it is cyclical and store sales are indeed much like the weather- seasonal. For a possible explanation for the decrease in sales during the warmer months, we must take into account that Rossmann is a drug store. During winter people get sick and the demand for cold and flu medication spikes, meanwhile during summer this demand falls.

Of note is that in the final model no weather variables contributed significantly to its performance (Figure 3.5c), max temperature on a given day was the 11th most important feature and the most important weather feature.

Key takeaways

- Rossmann performs the worst during warmer months.
- Generally, sales follow an inverse trend to weather.
- Weather follows a predictable trend with time of year.
- Thus, the effects of temperature variables are captured in the date variable as well.

3.6 Customer EDA

Purpose and Scope

This analysis aims to explore the relationship between customer - related factors (including promotional activities, time trends, store types, etc.) and the sales performance of Rossmann stores, as well as the close correlation between customer numbers and sales. The dataset covers multiple dimensions of Rossmann stores, such as daily sales, customer numbers, promotion indicators, date information, and store types, and spans a wide range of operational scenarios to gain an in - depth understanding of customer behavior and the impact of various factors on sales results.

Close Relationship between Customers and Sales

- **Customer Numbers Drive Sales:** There is a strong positive correlation between customer numbers and sales. On days with a large number of customers (such as promotional days, weekends, and Type B stores), sales are always higher. For example, promotional activities can increase the number of customers by dozens or even hundreds, directly leading to a significant increase in sales.
- **Synergistic Effect of Time and Promotion:** The number of customers fluctuates with time (seasonality, weekly cycle) and promotional activities, and sales closely follow these fluctuations — when the number of customers increases due to seasons or promotions, sales increase accordingly; when the number of customers decreases (such as on non - promotional working days), sales decrease.
- **Intermediary Role of Store Type:** Different store types attract different numbers of customers, which in turn affect sales. Type B stores attract the most customers and have the highest sales, while other types (A, C, D) have fewer customers and lower sales, indicating that store types indirectly affect sales by shaping customer traffic.

Recommendations

To optimize operations and revenue, Rossmann could:

1. **Use Promotional Activities Strategically:**
 - Focus on conducting promotions during seasonal peak periods such as summer and December, and amplify the effect by combining promotional momentum and seasonal demand.
 - Customize promotional efforts by month based on historical "sales lift" data (for example, increase promotional investment when the increase is the highest in June).
2. **Allocate Resources Based on Store Type ROI:**
 - Invest more resources (marketing, inventory, personnel) in Type B stores to further strengthen their leading position.
 - For poorly performing store types (Type C and D), improve customer traffic and sales through targeted promotions and customer loyalty programs, and evaluate the return on investment of converting some of these stores to the Type B model.
3. **Enhance Customer - Centric Analysis:**
 - Continuously analyse customer behavior under different dimensions (promotion, time, store type), identify subtle trends, and dynamically optimize strategies.
 - Use predictive models to forecast the number of customers and proactively adjust operations (such as preparing goods in advance during peak hours).

4. Predictive Models:

Predicting Customers

Figure 4.0a showed how the customers variable is a great estimate for sales, as the customers line shadows sales throughout the entire two year data period. Additionally using a correlation heatmap (Figure 4.0b), we see how customers are the most important variable to impact sales, with a correlation value of 0.9. Unfortunately however, when forecasting sales four weeks in advance, the customer's variable is unknown. So, we predicted customers and used that prediction for customers in the final model.

We took two different approaches in predicting customers, firstly we used the rolling mean. Here we used a rolling window of 28 days to create a value for customers on each forecasting day (each day in the four weeks ahead that we would be predicting sales for). Using this value and RandomForestRegressor as the model we were able to predict sales with a root mean squared percentage error (RMSPE) of 17%, testing R^2 of 0.87 and a training R^2 of 0.98. Here the much larger training value suggests the data is overfit, thus the lower testing score is unsurprising.

Our second approach to predict customers used RandomForestRegressor to predict a value for customers. Here customers and sales were dropped from the final dataset, sales was ignored and the target to predict was set to customers. We then split the data into training and testing sets and fit it using random forest. Here we had a RMSPE of 13.9%, testing R^2 of 0.95% and training R^2 of 0.98.

Unfortunately both approaches were overfit on the training data (which is undesirable as you want the model to be applied to new data and perform similarly), however as the random forest model produced a lower error we chose that going forward.

4.1 Random Forest

Description

Rossmann store sales are a continuous target, thus RandomForestRegressor was used for all random forest predictions rather than RandomForestClassifier. RandomForestRegressor works by using a collection of decision trees, then when predicting values (in this case sales), it averages the predictions from each individual tree. Importantly each tree is training independently on a different set of data with a random subset of features. Overall, random forest is useful for high dimensional data (e.g non linear) which this dataset is (see again figure 4.0a clearly sales is non linear), hence why random forest was chosen as a model to forecast sales.

Results

As previously mentioned, the customers variable that was given to use in the dataset cannot be used to forecast sales four weeks ahead of time, however to demonstrate the importance of customers random forest was run with the predicted vs actual customers. When predicting customers, random forest produced an RSMPE of 11.6%, R^2 test of 0.94 and R^2 train of 0.99, clearly very overfit and generally unsuitable for unseen data. However when using the actual customer value on a given day we could predict sales with an error of 6.8 %, with an R^2 test and train of 0.95. These results demonstrate the effectiveness of using random forest to predict sales as well as the importance of an effective prediction for customers.

4.2 Lasso Regression

Description

Lasso regression is quite different to the other models discussed so far and it was used to see how effective a linear approximation of sales would be. Again since sales is continuous the lasso algorithm works as a regression method not a classifier. Lasso, which stands for Least Absolute Shrinkage and Selection Operator, fits a linear model that minimises prediction error by applying a penalty to the absolute value of the coefficients (the variables), here it can shrink unimportant features to zero. Thus, the lasso model also demonstrates which features are most important in predicting sales (as all others will be set (or shrunk close to) zero).

We suspected that many of the variables would be unhelpful in the final model so using lasso regression allowed us to confirm our suspicions early whilst also seeing how a refined linear regression model would perform.

Importantly Lasso Regression uses an 'alpha' value to decide the penalty when shrinking values. This alpha value is a hyperparameter, thus to ensure the best performance of Lasso Regression we performed cross validation with five folds which gave us an optimal alpha of 2.84.

Results

Using the optimal alpha we found a RMSPE of 17.2 with R^2 values for both test and train at 0.878. Notably, this model used the real values for customers, so its prediction was poor compared to that of Random Forest or XG Boost for example. A possible explanation for this is simply due to the non linear nature of the sales data.

Additionally Lasso Regression highlighted which features were most important to the predictive model (Figure 4.2a). Here we see how most of the weather variables are inconsequential and how customers dominate other variables.

4.3 XG Boost

Description

The XGBoost model was implemented to increase prediction accuracy and deal with overfitting within the Rossmann sales forecasting task. XGBoost is a gradient boosting algorithm which builds an ensemble of decision trees where each next tree corrects the mistakes of the one before. This iterative approach enables the model to represent complex, non linear relationships between variables such as store characteristics, promotions, customer activity, competition and weather conditions.

Categorical variables were encoded using label encoding, and the target variable was log transformed to stabilise variance and decrease skewness. The model was trained on 80% of the data and tested on the remaining 20% with early stopping applied after 50 iterations to prevent overfitting. The chosen configurations consisted of 1000 estimators, a learning rate of 0.05 and a maximum depth of trees as 6. Model performance was evaluated using both Root Mean Squared Error (RMSE) and Root Mean Squared Percentage Error (RMSPE) to ensure comparability across stores with different sales numbers.

Results

The XGBoost model produced a training RMSE of 605.6 and a validation RMSE of 799.7, which were also represented by an R squared score of 0.67 and 0.940 respectively. The small difference between training and testing results show that the model generalised well without significant overfitting. When assessed using RMSPE, the testing score was about 0.104 which confirmed stable predictive accuracy across the dataset.

Feature importance analysis (Figure 4.3a) showed that the number of customers were the most important predictor of sales, with StoreType_d, StoreType_b and CompetitionDistance following. Other important variables that had an impact included State, Assortment and time based variables. Weather variables were also found to be one of the top features, but they had lower importance comparatively, suggesting that environmental factors have a small but tangible impact on sales performance.

In the future, the model could be improved using hyperparameter tuning, cross validation and testing additional external variables like regional economic indicators or consumer sentiment data. This would improve accuracy and ensure even stronger generalisation to unseen conditions.

4.4 Linear Regression

Description

Linear Regression was implemented to test whether a simple linear relationship between the input variables and store sales could explain a substantial portion of the variance in the data. This model assumes that the relationship between sales and its predictors is linear, which makes it interpretable and efficient to train. Linear regression also provides direct insights into the contribution of each variable (for instance, the customer and promotion coefficients) to overall sales performance.

Although linear regression does not handle nonlinear patterns or complex feature interactions, it serves as a strong baseline to evaluate whether additional model complexity (e.g. tree-based models) provides meaningful improvement.

Results

Linear Regression achieved an RMSPE of 26.4, a Test R^2 of 0.84, and a Train R^2 of 0.99. These results indicate that while the model fits the training data well, there is a degree of overfitting, as evidenced by the noticeable drop in the test R^2 .

Compared to regularised linear models such as LassoCV (RMSPE = 57.7) and RidgeCV (26.0), Linear Regression still performed competitively, showing that heavy regularisation was unnecessary for this dataset. Overall, the model provided a clear and interpretable baseline for evaluating more complex approaches tested later in the analysis.

4.5 Polynomial Regression via Support Vector Regression

Description

To investigate the predictive ability of customer-related factors on the sales of Rossmann stores, this analysis employs support vector regression (SVR) combined with a polynomial kernel function to construct a prediction model, aiming to capture the nonlinear relationships between sales and variables such as customer quantity, promotional activities, time characteristics, and store type.

- After feature engineering (one-hot encoding for categorical variables, filling of missing values, etc.), the data is standardized using MinMaxScaler to eliminate the influence of variable scale differences. The model training is conducted using a 80% training set and 20% validation set split, with the coefficient of determination (R^2) as the evaluation metric. The model configuration is SVR polynomial kernel (kernel='poly'), with parameters max_iter=100, C=100, and epsilon=0.3. However, during training, a "solver premature termination" message was displayed, indicating that the model did not fully converge within the iteration count.

Results

- In the scenario with customer fields, the R^2 values for the training set and validation set were -7.019 and -7.035 respectively; in the scenario without customer fields, the R^2 value for the validation set was -6.893, both of which were far below 0, indicating that even after data standardization, the polynomial kernel SVR still had severe underfitting and failed to effectively capture the variable relationships in the data.

4.6 Decision Tree

Description

A Decision Tree model was implemented to capture potential nonlinear relationships and interactions between predictors such as promotions, customers, and store type. Unlike Linear Regression, which assumes a straight-line relationship, Decision Trees recursively split the data into smaller, more homogeneous groups based on the most informative variables. This makes them capable of handling complex feature interactions and nonlinearity while remaining relatively interpretable.

However, Decision Trees are prone to overfitting, especially when the tree grows too deep, as they tend to memorize training patterns rather than generalize to unseen data. Proper tuning of hyperparameters such as maximum depth and minimum samples per split is therefore essential.

Results

The Decision Tree model achieved an RMSPE of 22.1, a Test R^2 of 0.93, and a Train R^2 of 1.00. These metrics indicate an excellent fit on the training data but suggest possible overfitting due to the perfect training performance.

Despite this, the model performed slightly better on the test data than Linear Regression, implying that some nonlinear interactions—particularly between promotions and customer traffic—were successfully captured. The results show that promotions continued to have a strong positive effect on sales, especially when combined with higher customer volumes.

Overall, while the Decision Tree offered improved predictive accuracy and captured more complex relationships, it lacked the simplicity and generalizability of Linear Regression, reinforcing the importance of balancing interpretability and performance in model selection.

4.7 Logistic Regression

Description

Logistic Regression was implemented to identify the factors that influence the strongest on whether a Rossmann store achieves high daily sales. Unlike models predicting continuous sales values, Logistic Regression estimates the probability of sales exceeding a defined threshold ($\text{HighSales} = 1$). It provides interpretable coefficients that indicate how each variable contributes to the likelihood of high sales. The model used standardised numerical inputs and one-hot-encoded categorical variables to ensure balanced feature scaling.

Results

The model achieved an accuracy of around 84%, showing reliable performance with limited overfitting. The coefficients revealed that promotions had the largest positive effect, confirming their significant role in driving sales. Monday and Friday indicators also had strong positive weights, reflecting the weekly demand cycle. Competition distance showed a mild negative impact, while state holidays contributed positively but varied by type.

Overall, Logistic Regression validated earlier EDA findings — that time-based and promotional factors are key drivers of sales, while competition exerts a smaller influence. The model's interpretability makes it valuable for understanding sales behaviour and confirming feature importance for use in more complex ensemble models.

4.8 Final Model (Stacked Ensemble Method)

Description

In an attempt to improve the RMSPE and reduce overfitting, we used a stack ensemble method. The idea here is that each method captures different relationships in the data, thus by using the ensemble of all of them one final model can be trained to predict sales more confidently than any individual method could.

See Figure 4.8a for a visual diagram of the ensemble. The methods used in the training models were XG Boost, Random Forest Regressor, Lasso Regression, Polynomial Regression (of degree two and three) as well as SVM (Support Vector Machines).

Results

We tested each of these methods as the potential final decision method. The results of the different methods being used as the final estimator can be seen in Figure 4.8b. Although the best model by RMSPE was Random Forest we chose XG Boost due to its lower training R^2 . This lower R^2 suggests it is less overfit to the training data and would in future perform a better estimation for unseen data.

Finally Figure 4.8c lists the most important features in the prediction.

5. Conclusion

What problem did we solve?

We set out to forecast Rossmann's daily store sales and identify levers managers can pull to improve performance under seasonality, promotions, store heterogeneity, and local competition.

What did the data say? Key takeaways.

- **Customers drive revenue.** Customer count is the strongest signal for sales; when unknown at forecast time, forecasting customers first materially improves sales forecasts.
- **Promotions work—but timing and store type matter.** Typical lifts are ~30–45%, largest in Apr–Jul and the biggest absolute gains in December. Types A/B respond best; C/D need tailored offers.
- **Store type is a structural advantage.** Type B leads on a per-store basis in both traffic and sales; replicating its playbook is an immediate upside for underperforming formats.
- **Competition effects are real but secondary.** Nearby and newly opened competitors depress sales most in the short run; effects attenuate over time and can be blunted with well-timed promotions.
- **Seasonality dominates weather.** Apparent weather effects are largely captured by date/season features; explicit weather variables added little incremental power.

What worked best for forecasting?

- Tree-based models handled non-linear patterns best. **XGBoost** with careful regularisation generalised well (test **RMSPE** $\approx$ **0.104**).
- **Random Forest** demonstrated the centrality of customers: using actual customers yielded **~6.8% error** (very strong), while using **predicted** customers increased error (**~11.6%**) and showed overfit in train vs test.
- Linear/Lasso baselines (e.g., **RMSPE** **~17%**) and polynomial/SVR variants underperformed on this problem's non-linear structure.
- A **stacked ensemble** with XGBoost as final estimator provided a robust balance of accuracy and generalisation relative to any single model.

So what should managers do?

By using a customer-first, promotion-aware, ensemble forecasting pipeline that is specific for each store, Rossmann can reliably anticipate demand, place inventory and labour where returns are highest, and dampen competitive shocks—turning store-day data into consistent, defensible gains in revenue and efficiency.

6. References

Dune_Dweller on Kaggle. (2025). *Rossmann Store Sales*. @Kaggle. State dataframe:

<https://www.kaggle.com/competitions/rossmann-store-sales/discussion/17048>

MCFG on kaggle. (2025). *Rossmann Store Sales*. @Kaggle. Weather dataframe:

<https://www.kaggle.com/competitions/rossmann-store-sales/discussion/17058#9707>

7. Appendix

Group Project Assessment for ADS1002							
Studio: Clayton Monday							
Group: 5							
		Angela	Felix	Haoyu	Jasmine	Nicole	Yanghan
Coordination	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	20	20	0	20	20	20
Research	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	20	20	0	20	20	20
Coding	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	17	20	12	17	17	17
Report Writing	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	17	17	15	17	17	17
Presentation Preparation	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	20	20	0	20	20	20
Attendance at Final Presentation	This entry should be 1 (present) or 0 (absent)	1	1	0	1	1	1

AI acknowledgement

I acknowledge the use of ChatGPT (<https://chat.openai.com/>) to improve the academic language of my work. On October 19th, parts of the essay were submitted with the prompt “Improve the grammar, vocabulary and write this in an academic tone.” The output was then modified to better reflect our own tone and style of writing.

Figure 3.1a: Average sales by weekday.

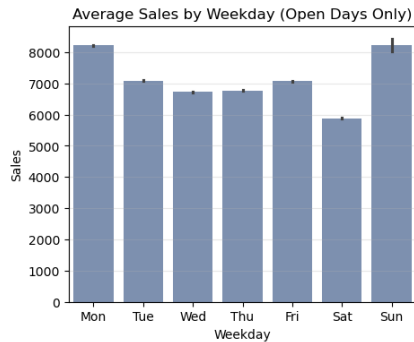


Figure 3.1b - c: Average sales by holiday type.

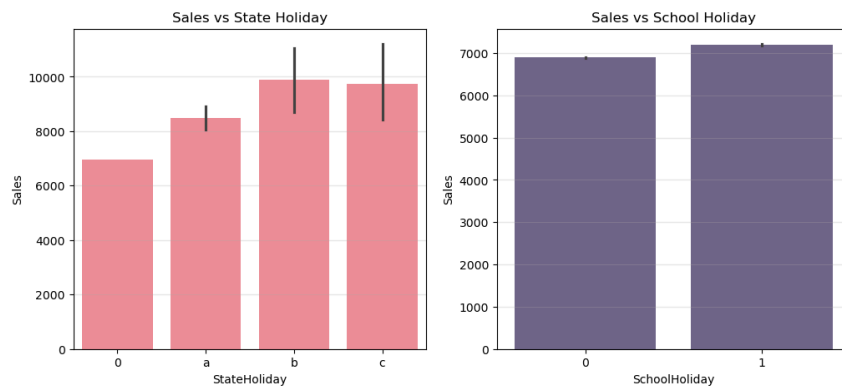


Figure 3.1d: Heatmap of average sales by weekday and state holiday.

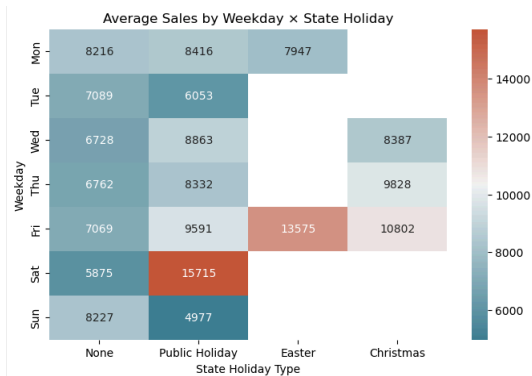


Figure 3.2 a: Sales distribution for promotional and non-promotional days.

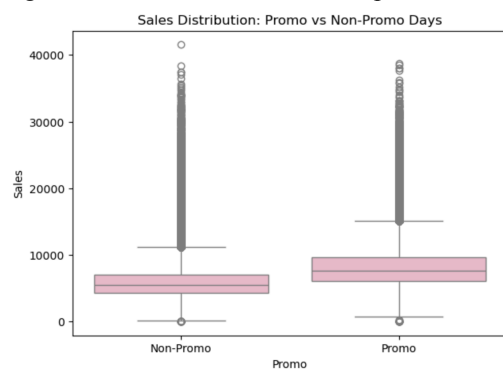


Figure 3.2 b - c: Monthly sales trend and comparison according to promotions.

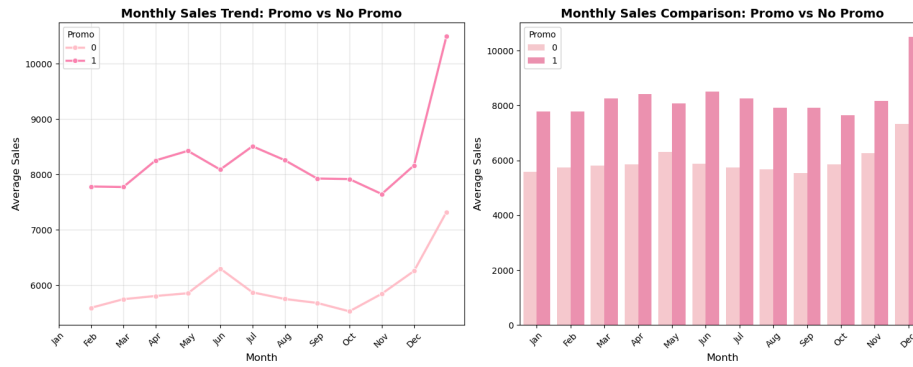


Figure 3.2 d: Promotional lift by month.

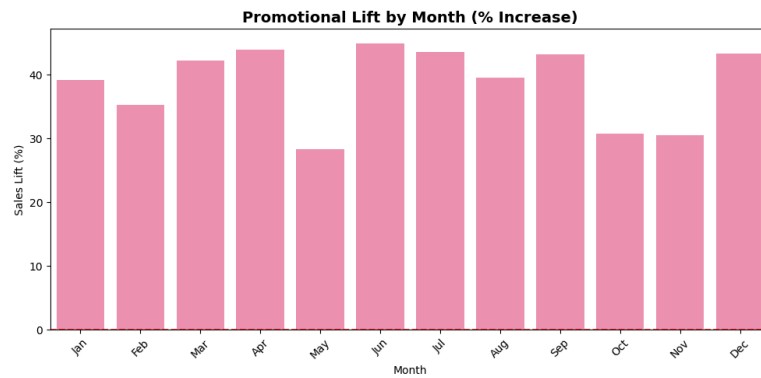


Table 3.2 a: Sales and promotional lift comparisons for specific months.

Month	No Promo (\$)	Promo (\$)	Lift (%)
4 (Apr)	5,855	8,426	43.9%
5 (May)	6,301	8,088	28.4%
6 (Jun)	5,872	8,509	44.9%
7 (Jul)	5,752	8,258	43.6%
12 (Dec)	11,073	15,623	41.1%

Figure 3.2 e: Average sales by store type and promotions.

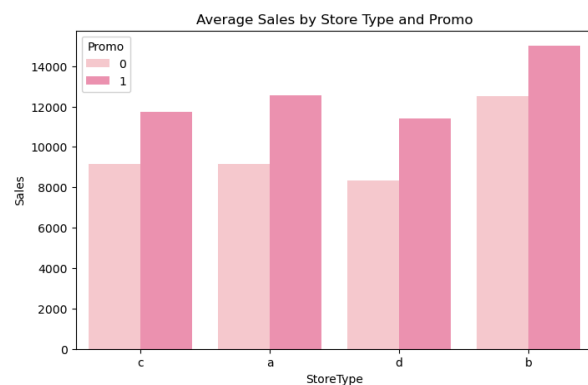


Figure 3.4a: Distribution of Competition Distance.

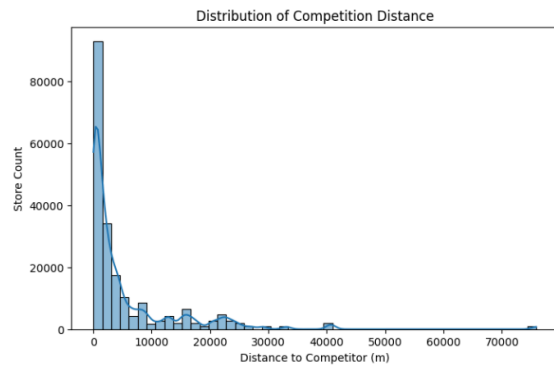


Figure 3.4b: Average Sales by Distance.

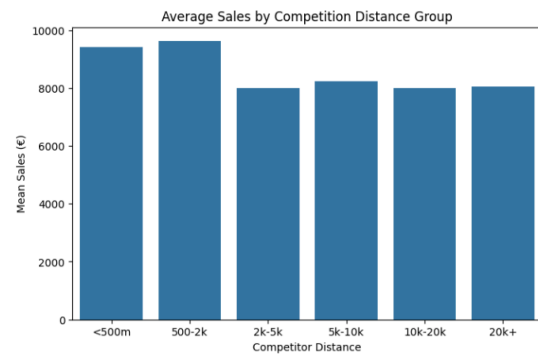


Figure 3.4c: Sales vs Competitor Age.

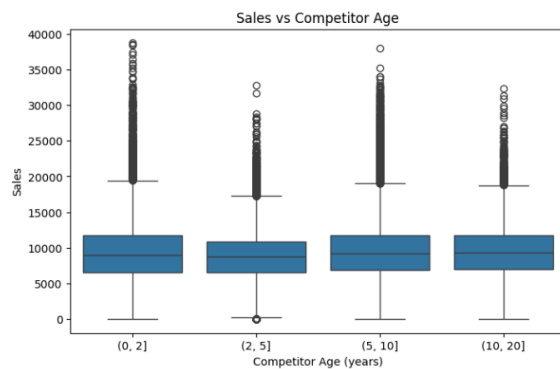


Figure 3.5a: Complete set of weather variables.

Date, Max_TemperatureC, Mean_TemperatureC, Min_TemperatureC, Dew_PointC, MeanDew_PointC, Min_DewpointC, Max_Humidity, Mean_Humidity, Min_Humidity, Max_Sea_Level_PressurehPa, Mean_Sea_Level_PressurehPa, Min_Sea_Level_PressurehPa, Max_VisibilityKm, Mean_VisibilityKm, Min_VisibilityKm, Max_Wind_SpeedKm_h, Mean_Wind_SpeedKm_h, Max_Gust_SpeedKm_h, Precipitationmm, CloudCover, Events, and WindDirDegrees.

Figure 3.5b: Graphing monthly weather and sales.

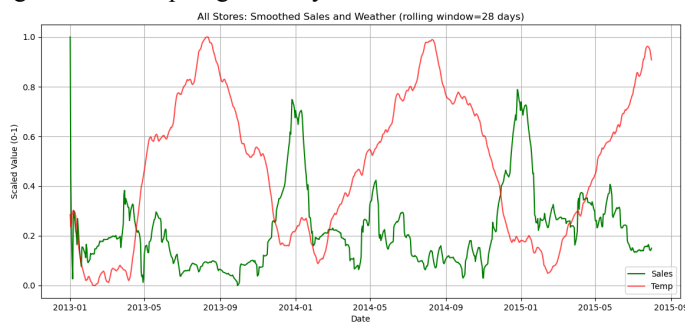


Figure 3.5c: Final model feature importance.

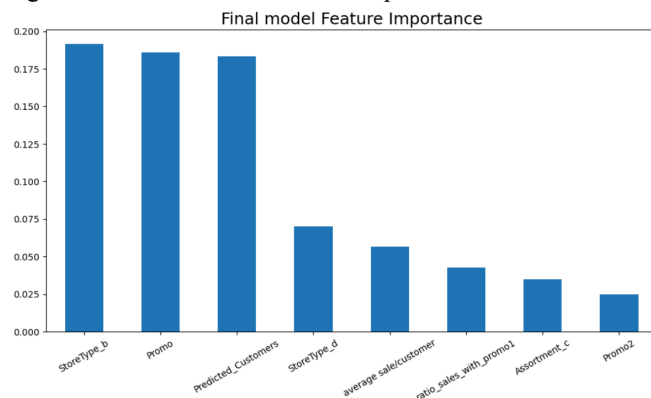


Figure 4.0a: Relationship between sales and customers.

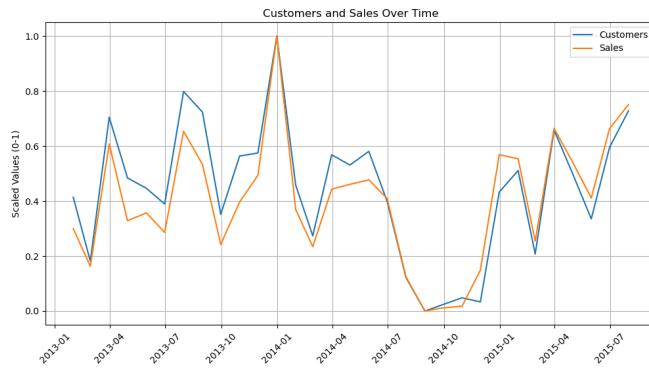


Figure 4.0b: Correlation heatmap for store variables.

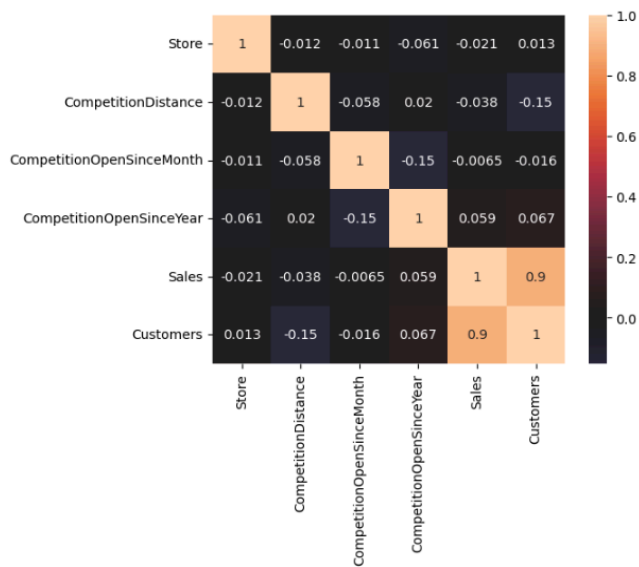


Figure 4.2a (see customers value at 3700)

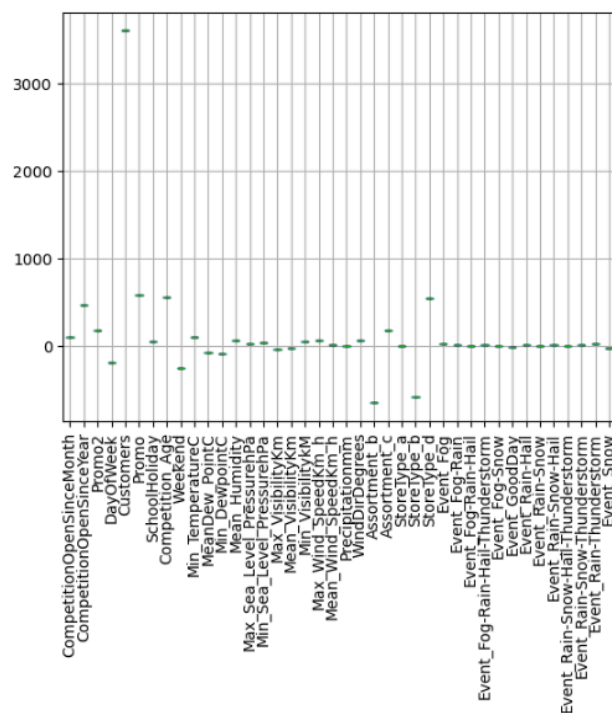
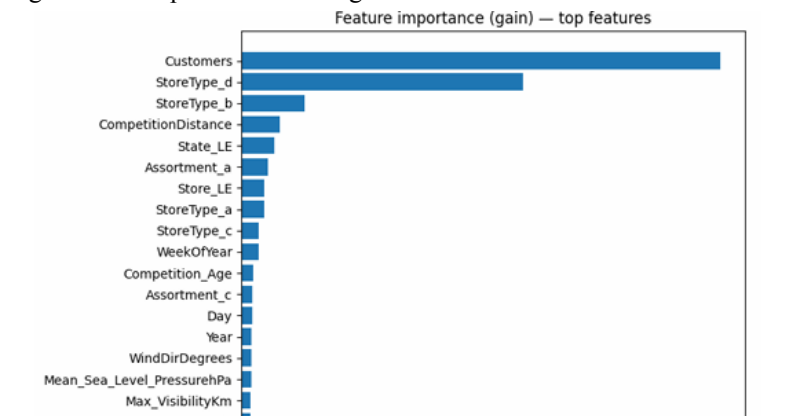


Figure 4.3a: Top features ranking.



Customers is the dominant feature, with StoreType_d and StoreType_b being the next most important predictors in the model according to XGBoost

Figure 4.8a: Stack ensemble method diagram.

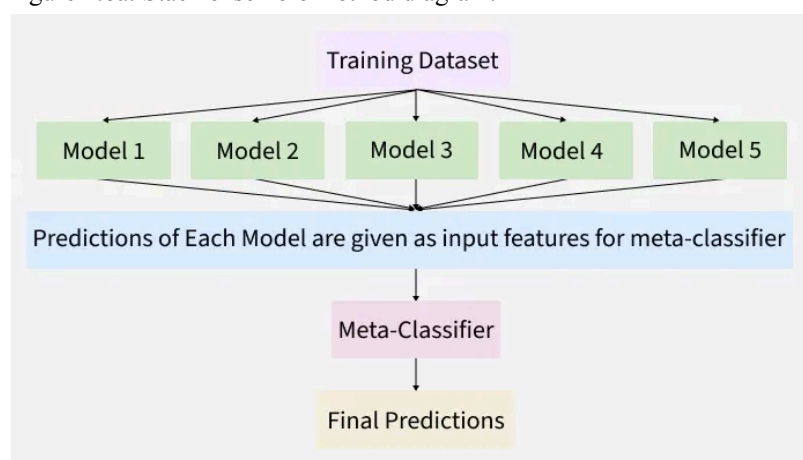


Figure 4.8b: Results for different meta.

Model	RMSPE	Test R ²	Train R ²
Random Forest	10.4	0.94	0.99
Linear Regression	26.4	0.84	0.99
XGBoost	10.9	0.94	0.97
SVR (RBF)	49.9	-0.05	-0.04
LassoCV	57.7	0.0	0.0
RidgeCV	26.0	0.84	0.99

Figure 4.8c: Final model feature importance.

StoreType_b	0.191460
Promo	0.185902
Predicted_Customers	0.183430
StoreType_d	0.070278
average sale/customer	0.056609
ratio_sales_with_promo1	0.042745
Assortment_c	0.034818
Promo2	0.024764
Assortment_a	0.024092
DayOfWeek	0.019212
StoreType_a	0.015009
Max_TemperatureC	0.012077
Min_Visibilitykm	0.011532
Mean_TemperatureC	0.009286
Store	0.007511